MTH2008 – Real Analysis

2026/27

Theorem

If $\lim_{x \to x_0} f(x)$ exists, then it is unique; that is, if

$$\lim_{x \to x_0} f(x) = L_1 \quad \text{and} \quad \lim_{x \to x_0} f(x) = L_2,$$

then $L_1 = L_2$.

Theorem

If

$$\lim_{x \to x_0} f(x) = L_1 \quad \text{and} \quad \lim_{x \to x_0} g(x) = L_2,$$

then

$$\lim_{x \to x_0} (f + g)(x) = L_1 + L_2,$$
$$\lim_{x \to x_0} (f - g)(x) = L_1 - L_2,$$
$$\lim_{x \to x_0} (fg)(x) = L_1 L_2,$$

and, if $L_2 \neq 0$,

$$\lim_{x \to x_0} \left(\frac{f}{g} \right)(x) = \frac{L_1}{L_2}.$$

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One-sided limits

Consider the fucntion $g : \mathbb{R} \to \mathbb{R}$ defined by

$$g(x) = \begin{cases} x + 1 & \text{if } x \geq 0, \\ x - 1 & \text{if } x < 0. \end{cases}$$

Observe that $g(x)$ does not approach a limit as x approaches 0. since for arbitrarily small but positive x, we have that $f(x)$ is *close* to 1 but for arbitrarily small but negative x, $f(x)$ is *close* to -1. This motivates our next definition.

Definition

(a) We say that $f(x)$ **approaches the left-hand limit L as x approaches x_0 from the left**, and write

$$\lim_{x \to x_0 -} f(x) = L,$$

if f is defined on some open interval (a, x_0) and, for each $\epsilon > 0$, there is a $\delta > 0$ such that, if $x_0 - \delta < x < x_0$, then

$$|f(x) - L| < \epsilon$$

(b) We say that $f(x)$ **approaches the right-hand limit L as x approaches x_0 from the right**, and write

$$\lim_{x \to x_0 +} f(x) = L,$$

if f is defined on some open interval (x_0, b) and, for each $\epsilon > 0$, there is a $\delta > 0$ such that, if $x_0 < x < x_0 + \delta$

$$|f(x) - L| < \epsilon.$$

Left- and right-hand limits are also called **one-sided limits**.

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Theorem

A function f has a limit at x_0 if and only if it has left- and right-hand limits at x_0, and they are equal. More specifically,

$$\lim_{x \to x_0} f(x) = L$$

if and only if

$$\lim_{x \to x_0-} f(x) = \lim_{x \to x_0+} f(x).$$

Recall:

Theorem

If $\lim_{x\to x_0} f(x)$ exists, then it is unique; that is, if

$$\lim_{x\to x_0} f(x) = L_1 \quad \text{and} \quad \lim_{x\to x_0} f(x) = L_2,$$

then $L_1 = L_2$.

Theorem

If

$$\lim_{x\to x_0} f(x) = L_1 \quad \text{and} \quad \lim_{x\to x_0} g(x) = L_2,$$

then

$$\begin{aligned}\lim_{x\to x_0} (f+g)(x) &= L_1 + L_2,\\ \lim_{x\to x_0} (f-g)(x) &= L_1 - L_2,\\ \lim_{x\to x_0} (fg)(x) &= L_1 L_2,\end{aligned}$$

and, if $L_2 \neq 0$,

$$\lim_{x\to x_0} \left(\frac{f}{g}\right)(x) = \frac{L_1}{L_2}.$$

Remark

With only minor modifications of their proofs (replacing the inequality $0 < |x - x_0| < \delta$ by $x_0 - \delta < x < x_0$ or $x_0 < x < x_0 + \delta$), it can be shown that the assertions of the above two theorems remain valid if "$\lim_{x\to x_0}$" is replaced by "$\lim_{x\to x_0-}$" (or "$\lim_{x\to x_0+}$") throughout.

Limits at $\pm\infty$

Limits and one-sided limits have to do with the behaviour of a function f near to some point of D – the domain of f. It is equally reasonable to study f for large positive values of x if D is unbounded above or for large negative values of x if D_f is unbounded below.

Definition

We say $f(x)$ **approaches the limit** L **as** x **approaches** ∞, and write

$$\lim_{x\to\infty} f(x) = L,$$

if f is defined on an interval (a, ∞) and, for each $\epsilon > 0$, there exists a real number β such that, for all $x \in D_f$ with $x > \beta$,

$$|f(x) - L| < \epsilon.$$

Similarly, $f(x)$ **approaches the limit** L **as** x **approaches** $-\infty$, written

$$\lim_{x\to-\infty} f(x) = L,$$

if f is defined on an interval $(-\infty, b)$ and, for each $\epsilon > 0$, there exists a real number α such that, for all $x \in D_f$ with $x < \alpha$,

$$|f(x) - L| < \epsilon.$$

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Infinite limits

The functions

$$f(x) = \frac{1}{x}, \quad g(x) = \frac{1}{x^2}, \quad p(x) = \sin\frac{1}{x},$$

and

$$q(x) = \frac{1}{x^2}\sin\frac{1}{x}$$

do not have limits, or even one-sided limits, at $x_0 = 0$. They fail to have limits in different ways:

- $f(x)$ increases beyond bound as x approaches 0 from the right and decreases beyond bound as x approaches 0 from the left;
- $g(x)$ increases beyond bound as x approaches zero;
- $p(x)$ oscillates with ever-increasing frequency as x approaches zero;
- $q(x)$ oscillates with ever-increasing amplitude and frequency as x approaches 0.

Definition

We say $f(x)$ **approaches ∞ as x approaches x_0 from the left**, and write

$$\lim_{x \to x_0-} f(x) = \infty,$$

if f is defined on an interval (a, x_0) and, for each real number M, there is a $\delta > 0$ such that for all $x \in D_f$ with $x_0 - \delta < x < x_0$

$$f(x) > M.$$

Throughout, $\lim_{x \to x_0} f(x)$ **exists** will mean that

$$\lim_{x \to x_0} f(x) = L,$$

where L is finite.

To leave open the possibility that $L = \pm\infty$, we will say that $\lim_{x \to x_0} f(x)$ **exists in the extended reals**. This convention also applies to one-sided limits and limits as x tends to $\pm\infty$.

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Monotonic functions

A function f is **non-decreasing** on an interval I if

$$f(x_1) \leq f(x_2),$$

whenever x_1 and x_2 are in I and $x_1 < x_2$. Similarly we say that f is **non-increasing** on I if

$$f(x_1) \geq f(x_2).$$

whenever x_1 and x_2 are in I and $x_1 < x_2$.

A function f is (**strictly**) **decreasing** on an interval I if

$$f(x_1) > f(x_2),$$

whenever x_1 and x_2 are in I and $x_1 < x_2$. Similarly, f is **increasing** on I if

$$f(x_1) < f(x_2).$$

whenever x_1 and x_2 are in I and $x_1 < x_2$.

In any of these four cases, we call f **monotonic** on I.

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Theorem

Suppose that f is monotonic on an interval (a, b) and define

$$\alpha = \inf_{a<x<b} f(x) \quad \text{and} \quad \beta = \sup_{a<x<b} f(x).$$

(a) If f is non-decreasing, then

$$\lim_{x\to a^+} f(x) = \alpha \quad \text{and} \quad \lim_{x\to b^-} f(x) = \beta$$

(b) If f is non-increasing, then

$$\lim_{x\to a^+} f(x) = \beta \quad \text{and} \quad \lim_{x\to b^-} f(x) = \alpha$$

(c) If $a < x_0 < b$, then

$$\lim_{x\to x_0^+} f(x) \quad \text{and} \quad \lim_{x\to x_0^-} f(x)$$

exist and are finite. Moreover,

$$\lim_{x\to x_0^-} f(x) \le f(x_0) \le \lim_{x\to x_0^+} f(x)$$

if f is non-decreasing, and

$$\lim_{x\to x_0^-} f(x) \ge f(x_0) \ge \lim_{x\to x_0^+} f(x)$$

if f is non-increasing.

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2.2 Continuity

The definitions of

$$\lim_{x \to x_0} f(x), \quad \lim_{x \to x_0^-} f(x), \quad \text{and} \quad \lim_{x \to x_0^+} f(x),$$

do not involve $f(x_0)$ or even require that it be defined. However, the case where $f(x_0)$ is defined and equal to one or more of these quantities is important and relates to the concept of continuity.

Definition

(a) We say that f is **continuous at** x_0 if f is defined on an open interval (a, b) containing x_0 and

$$\lim_{x \to x_0} f(x) = f(x_0).$$

(b) We say that f is **continuous from the left at** x_0 if f is defined on an open interval (a, x_0) and

$$\lim_{x \to x_0-} f(x) = f(x_0).$$

(c) We say that f is **continuous from the right at** x_0 if f is defined on an open interval (x_0, b) and

$$\lim_{x \to x_0+} f(x) = f(x_0).$$

Example

Determine at which points the function $f : [0, 2] \to \mathbb{R}$ defined, for $x \in [0, 2]$, by

$$f(x) = \begin{cases} x^2, & \text{if } 0 \leq x < 1, \\ x + 1, & \text{if } 1 \leq x \leq 2. \end{cases}$$

is continuous.

Definition

- A function f is **continuous on an open interval** (a, b) if it is continuous at every point in (a, b).
- If a function f is continuous on (a, b) and

$$\lim_{x \to b^-} f(x) = f(b),$$

then we say that f is **continuous on** $(a, b]$.

- If a function f is continuous on (a, b) and

$$\lim_{x \to a^+} f(x) = f(a),$$

then we say that f is **continuous on** $[a, b)$.

- If f is continuous on (a, b) and

$$\lim_{x \to b^-} f(x) = f(b) \quad \text{and} \quad \lim_{x \to a^+} f(x) = f(a),$$

then we say that f **is continuous on** $[a, b]$.

- More generally, if f is a function with domain D, and if S is a subset of D, consisting of finitely or infinitely many disjoint intervals, then f is **continuous on** S if f is continuous on every interval in S.

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Definition

A function f is **piecewise continuous** on $[a, b]$ if

(a) $\lim_{x \to x_0^+} f(x)$ exists for all x_0 in $[a, b)$;

(b) $\lim_{x \to x_0^-} f(x)$ exists for all x_0 in $(a, b]$;

(c) $\lim_{x \to x_0^-} f(x) = \lim_{x \to x_0^+} f(x)$ for all but finitely many $x_0 \in (a, b)$.

If (c) fails to hold at some x_0 in (a, b), f is said to have a **jump discontinuity at** x_0.

Also, f has a **jump discontinuity at** a if

$$\lim_{x \to a^+} f(x) \neq f(a)$$

and a **jump discontinuity at** b if

$$\lim_{x \to b^-} f(x) \neq f(b).$$

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Theorem

If f and g are continuous on a set S, then so are $f + g$, $f - g$, and fg. In addition, f/g is continuous at each x_0 in S such that $g(x_0) \neq 0$.

Removable discontinuities

Let f be defined on a deleted neighborhood of x_0 and discontinuous (perhaps even undefined) at x_0. We say that f has a **removable discontinuity** at x_0 if $\lim_{x \to x_0} f(x)$ exists.

In this case, the function

$$g(x) = \begin{cases} f(x) & \text{if } x \in D \text{ and } x \neq x_0, \\ \lim_{x \to x_0} f(x) & \text{if } x = x_0, \end{cases}$$

is continuous at x_0.

Example

Composite functions

We have seen that the investigation of limits and continuity can be simplified by regarding a given function as the result of addition, subtraction, multiplication, and division of simpler functions. Another operation useful in this connection is **composition** of functions; that is, the substitution of one function into another.

Definition

Suppose that f and g are functions with domains D_f and D_g. If D_g has a nonempty subset T such that $g(x) \in D_f$ whenever $x \in T$, then the **composite function** $f \circ g$ is defined on T by

$$(f \circ g)(x) = f(g(x)).$$

Example

Theorem

Suppose that g is continuous at x_0, $g(x_0)$ is an interior point of D_f, and f is continuous at $g(x_0)$. Then $f \circ g$ is continuous at x_0.

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Bounded functions

- A function f is **bounded below** on a set S if there exists a real number m such that, for all $x \in S$,

$$f(x) \geq m.$$

In this case, the set

$$V = \{f(x) \mid x \in S\}$$

has an infimum α, and we write

$$\alpha = \inf_{x \in S} f(x).$$

If there exists a point $x_1 \in S$ such that $f(x_1) = \alpha$, we say that α is the **minimum of f on S**, and write

$$\alpha = \min_{x \in S} f(x).$$

- Similarly, f is **bounded above on S** if there exists a real number M such that, for all x in S,

$$f(x) \leq M.$$

In this case, V has a supremum β, and we write

$$\beta = \sup_{x \in S} f(x).$$

If there exists $x_2 \in S$ such that $f(x_2) = \beta$, then we say that β is the **maximum of f on S**, and write

$$\beta = \max_{x \in S} f(x).$$

- If f is bounded above and below on a set S, we say that f is **bounded** on S.

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Theorem

If f is continuous on a finite closed interval $[a, b]$, then f is bounded on $[a, b]$.

Theorem

Suppose that f is continuous on a finite closed interval $[a, b]$. If

$$\alpha = \inf_{a \le x \le b} f(x) \quad \text{and} \quad \beta = \sup_{a \le x \le b} f(x),$$

then α and β are respectively the minimum and maximum of f on $[a, b]$.

Namely, there exist $x_1, x_2 \in [a, b]$ with

$$f(x_1) = \alpha \quad \text{and} \quad f(x_2) = \beta.$$